

STORM: Spatially Aware H&E-ST Foundation Model

A concise reading map linking the biological question, representation design, evidence, and reusable ideas.

Paper	A Multimodal Foundation Model of Spatial Transcriptomics and Histology for Biological Discovery and Clinical Prediction		
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Authors	Jinxi Xiang, Siyu Hou, Yuchen Li, Ryan Quinton, Xiaoming Zhang, et al.		
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Reader / Date	Ye Zhang		May 19, 2026

Table 1: At a glance summary of the STORM paper.

Aspect	Reading note
Question	Can a single spatially aware foundation model bridge routine H&E images and spatial transcriptomics for biological discovery and clinical prediction?
Data	1.2 million spatially resolved transcriptomic profiles with matched histology from 632 tissue sections across 18 organs and multiple ST platforms.
Model	STORM uses spot-level H&E/ST encoders plus a spatial encoder that integrates local morphology, gene expression, and neighborhood context through masked multimodal pretraining.
Output	Spatial domains, virtual ST from H&E, multimodal prognosis prediction, immunotherapy response prediction, and multi-level biological interpretation.
Main claim	Spatially aware H&E-ST pretraining improves both molecular reconstruction and clinical prediction beyond H&E-only models, ST-only models, and prior H&E-ST foundation models.

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1 One-sentence summary

SUMMARY 1.1 Core summary

STORM is a spatially aware multimodal foundation model that learns joint representations from histology, spatial transcriptomics, and local tissue neighborhoods, enabling spatial domain discovery, H&E-based virtual ST prediction, and clinically useful outcome prediction from routine pathology slides.

2 Why this paper matters

This paper sits at the intersection of three trends: pathology foundation models, spatial transcriptomics, and clinically deployable virtual omics. The core idea is attractive because true spatial transcriptomics is expensive and low-throughput, while H&E slides are cheap, routine, and widely available. If H&E can be translated into spatially resolved molecular information, many downstream spatial biology analyses could become much more scalable.

Compared with OmiCLIP and Multi-Embed, STORM emphasizes **spatial context**. It is not only aligning one image patch to one gene-expression vector; it explicitly models a local neighborhood of spots. This is important because tissue biology is spatially organized: tumor-stroma interfaces, TLS regions, hypoxic niches, and invasive fronts are not single-spot phenomena.

INSIGHT 2.1 Key takeaways

The key conceptual contribution is the shift from patch-level H&E-ST alignment to **spatially aware H&E-ST representation learning**. STORM tries to learn not only what a spot looks like and expresses, but also how that spot sits inside a local tissue architecture.

3 Problem formulation

The paper addresses three connected tasks:

- ▶ **Spatial domain discovery:** use H&E, ST, or both to recover biologically coherent tissue compartments.
- ▶ **Virtual ST prediction:** predict spatial transcriptomic profiles directly from H&E images.
- ▶ **Clinical prediction:** use H&E-derived virtual ST plus morphology for prognosis and immunotherapy outcome prediction.

The model is trained on paired H&E-ST data and then reused for downstream settings where true ST may not be available. This makes the framework clinically motivated: expensive ST is used during pretraining, while deployment can rely on routine H&E.

ASSUMPTION 3.1 Core modeling assumption

The paper assumes that histology contains recoverable molecular information, but that this information is best decoded when local spatial context is modeled. A single isolated patch may be ambiguous; a neighborhood of patches can better reveal tissue compartments and microenvironmental organization.

4 Method breakdown

4.1 Data scale and model architecture

STORM is pretrained on a large pan-tissue H&E-ST dataset: 632 paired ST-H&E slides across 18 organs and approximately 1.2 million high-resolution tissue spots. The input data include multiple ST platforms such as Visium, ST, Xenium, and Visium HD.

The architecture has a two-level hierarchy:

1. **Spot-level encoders:** an H&E encoder extracts morphology features, and an ST encoder extracts gene-expression features.
2. **Spatial encoder:** a neighborhood of spots is processed jointly with spatial positions, allowing the model to capture local tissue organization.
3. **Masked multimodal pretraining:** H&E/ST spots are randomly masked and reconstructed, forcing the model to infer missing biological information from surrounding morphology and molecular context.

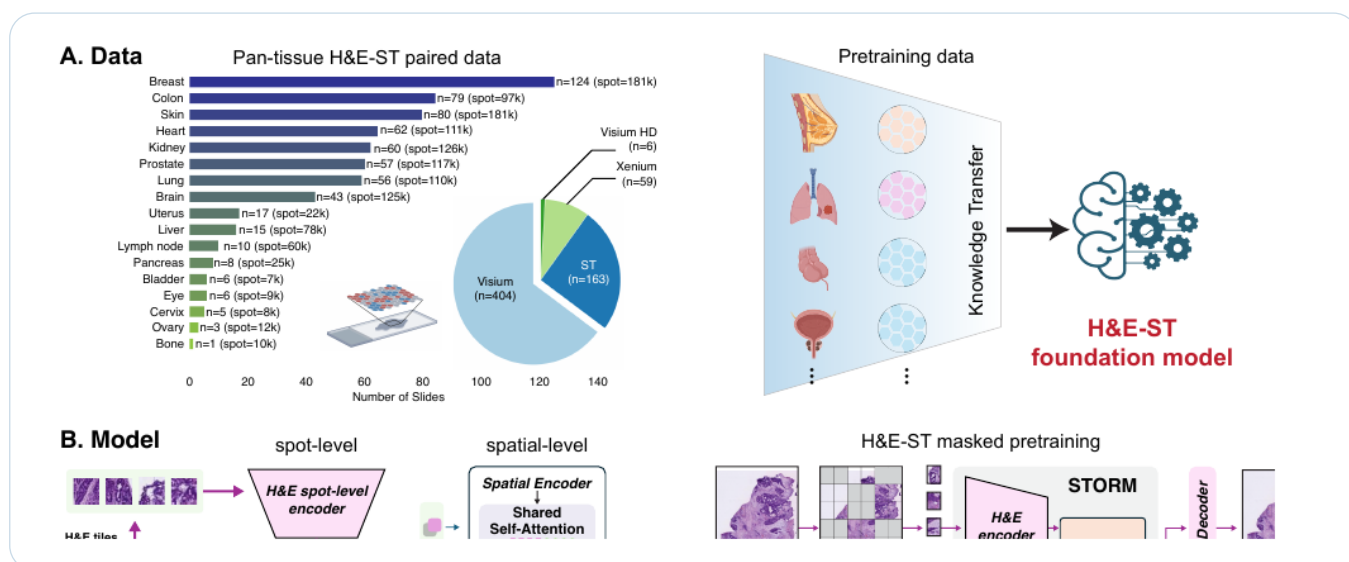


Figure 1: Figure 1A-B of the paper. STORM uses pan-tissue paired H&E-ST data and a hierarchical architecture combining spot-level encoders with a spatial encoder. The masked pretraining objective reconstructs missing H&E and ST information from local tissue context.

METHOD 4.1 Method breakdown

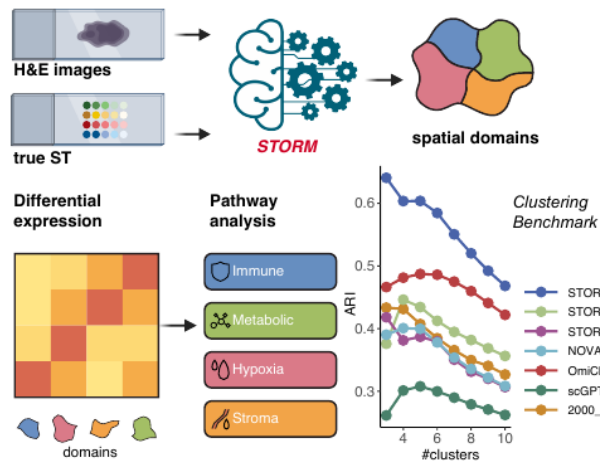
The important design choice is the spatial encoder. Many H&E-to-ST methods treat each spot independently, but STORM treats the tissue as a structured neighborhood. This makes the model closer to spatial biology than to ordinary patch-level regression.

4.2 Application landscape

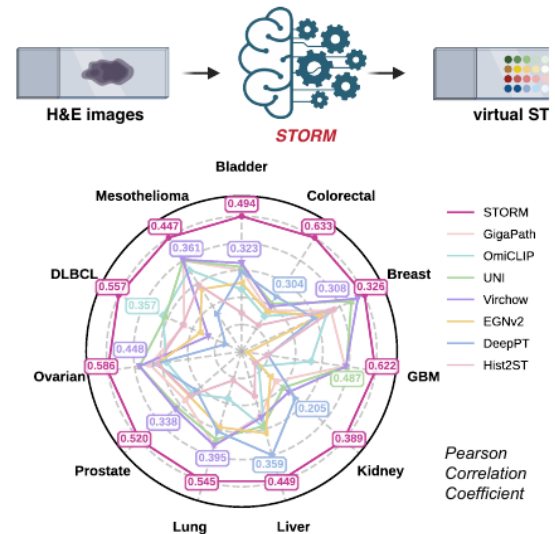
The paper evaluates STORM on spatial biology and clinical tasks: spatial domain discovery, virtual ST generation, prognosis prediction, and immunotherapy response prediction.

C. Applications

(1) spatial domain discovery with H&E-ST inputs



(2) predicting virtual ST from H&E images



(3) clinical outcome predictions

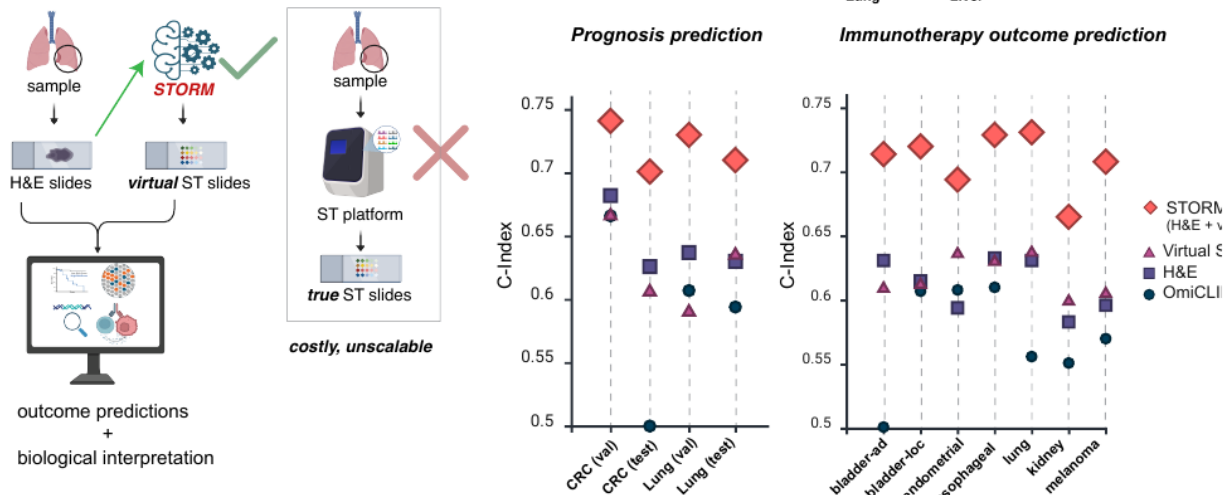


Figure 2: Figure 1C of the paper. STORM is evaluated as a reusable foundation model for spatial domain discovery, virtual ST prediction from H&E, and clinical outcome prediction using H&E plus generated virtual ST.

FIGURE NOTE 4.2 Why this overview is useful

The paper is not built around one narrow benchmark. The same representation is tested for biological discovery and clinical prediction, which makes the claim closer to a foundation-model story than a task-specific H&E-to-gene model.

5 Main experimental results

EDITORIAL NOTE 5.1 How this section is organized

Following the previous reading-note style, I split the experimental evidence into separate subsections. Each subsection contains a cropped result figure and a short interpretation, so the experimental claims are visually grounded rather than described only in text.

5.1 Spatial domain discovery: benchmark performance

The first major experiment evaluates multimodal spatial domain discovery using kidney cancer and lung cancer Visium datasets with H&E images and pathologist annotations. STORM is compared with 2,000 HVGs, OmiCLIP, NOVAE, and scGPT-spatial. In the UMAP and spatial maps, STORM better separates tumor, normal, TLS, and immune infiltration regions.

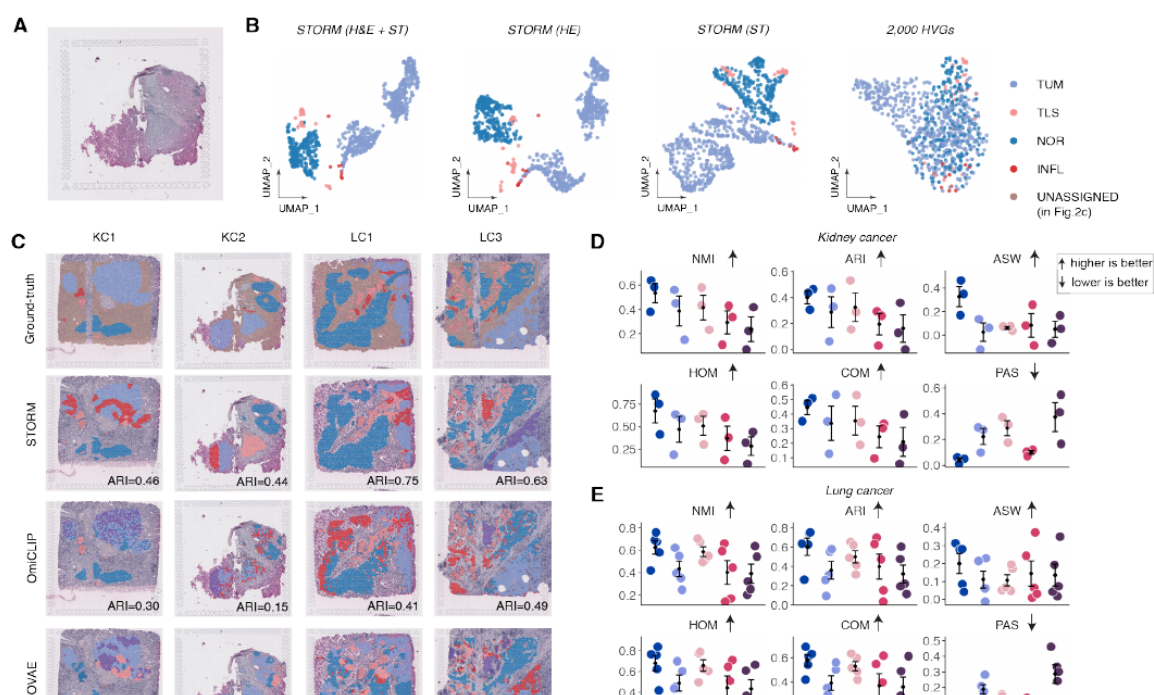


Figure 3: Figure 2A-F of the paper. STORM improves spatial domain discovery compared with HVG-based clustering and foundation-model baselines. The performance is evaluated by NMI, ARI, homogeneity, compactness, and abnormal spot percentage across clustering resolutions.

FIGURE NOTE 5.2 Interpretation

This experiment supports the idea that spatially aware H&E-ST embeddings preserve tissue-compartment information better than purely transcriptomic or less spatially contextualized foundation embeddings. The robustness across the number of clusters is also important because domain discovery can be sensitive to resolution choice.

5.2 Biological discovery from spatial domains

The next part of Figure 2 asks whether the discovered domains are biologically meaningful. In KC2, STORM splits a coarse tumor region into three tumor subdomains. Domain 3 is enriched for genes and pathways related to metabolism, inflammation, hypoxia, EMT, and tumor aggressiveness. The marker signature is further validated

in TCGA-KIRC, where high expression corresponds to poorer prognosis.

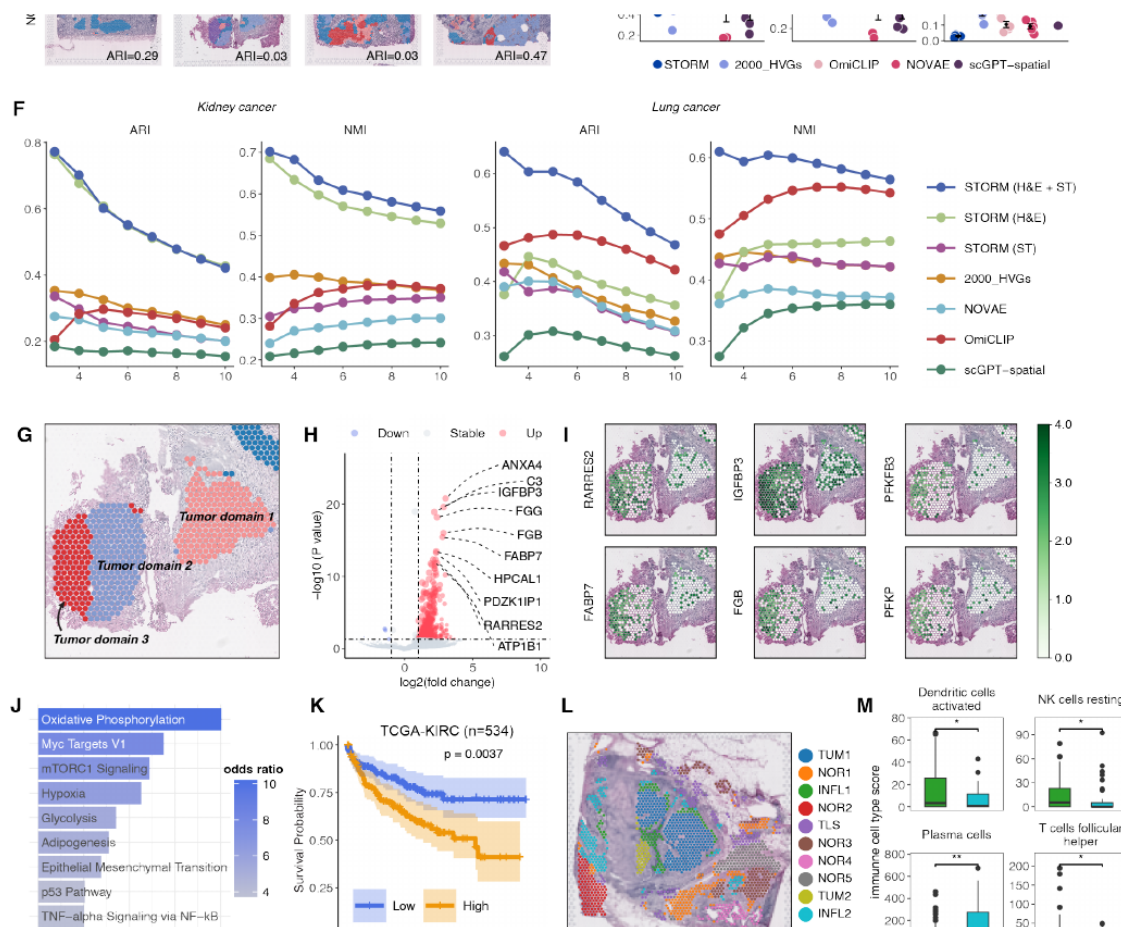


Figure 4: Figure 2G-M of the paper. STORM identifies biologically distinct tumor and immune subdomains. Domain 3 in kidney cancer shows aggressive metabolic and inflammatory programs, and fine-resolution clustering separates TLS and infiltrate subtypes.

FIGURE NOTE 5.3 Why this result matters

This is the strongest biological-discovery example. The model does not only improve clustering metrics; it uncovers a tumor subregion with coherent gene programs and prognostic relevance. This makes the spatial domain result more convincing than a purely visual clustering comparison.

5.3 Virtual ST prediction from H&E: benchmark across tumor types

The second major task is predicting spatial transcriptomics directly from H&E images. STORM is compared with task-specific H&E-to-ST models, pathology foundation models, and OmiCLIP. Across the displayed datasets, STORM achieves the highest gene-wise PCC for 500 HVGs.

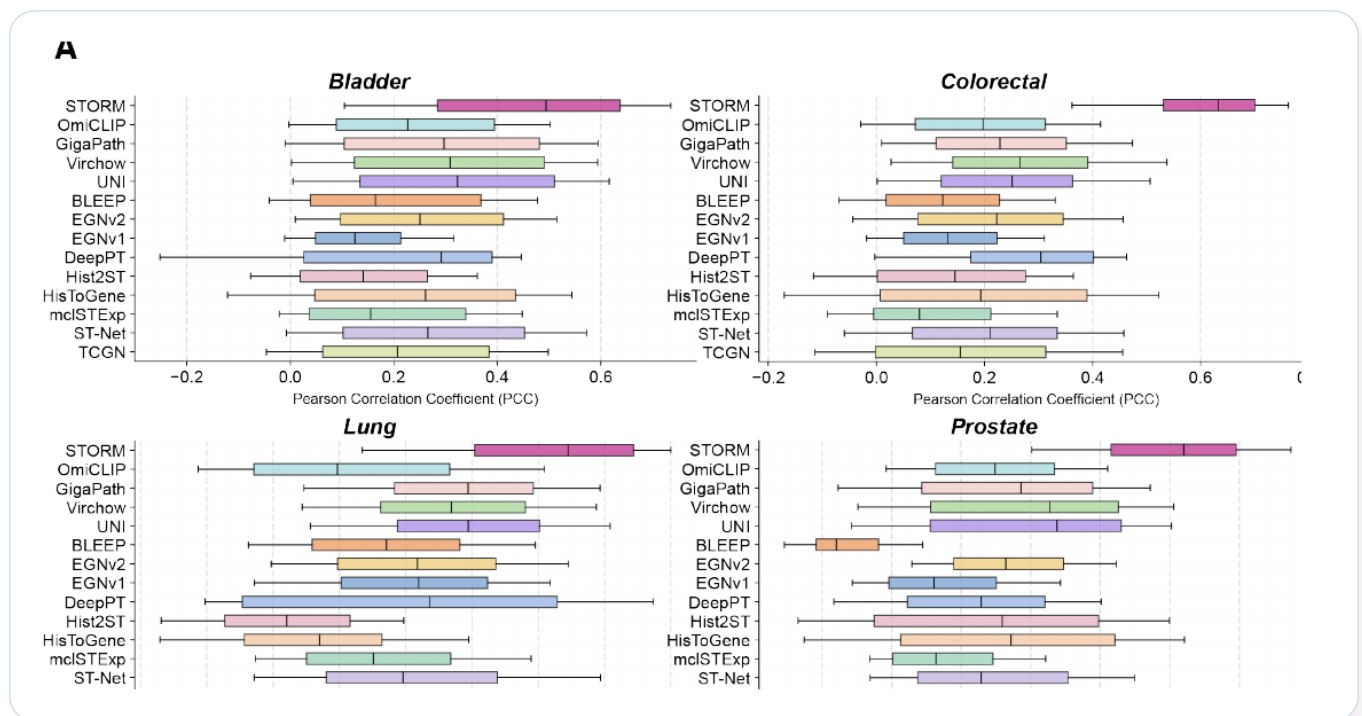


Figure 5: Figure 3A of the paper. STORM outperforms competing H&E-to-ST, pathology foundation, and H&E-ST foundation models in gene-expression prediction for bladder, colorectal, lung, and prostate datasets.

FIGURE NOTE 5.4 Key reading

This panel is central to the paper. It shows that spatially aware multimodal pretraining is not merely useful for clustering true ST data; it also improves the harder task of generating virtual ST from H&E alone.

5.4 Top-gene prediction and robustness to gene-panel size

The paper further evaluates whole-transcriptome prediction by ranking genes according to PCC. STORM remains ahead of OmiCLIP, Virchow, and DeepPT across eight tissues for the top 500 genes. In the colorectal example, the advantage remains visible across gene-panel sizes from 100 to 2,000 genes.

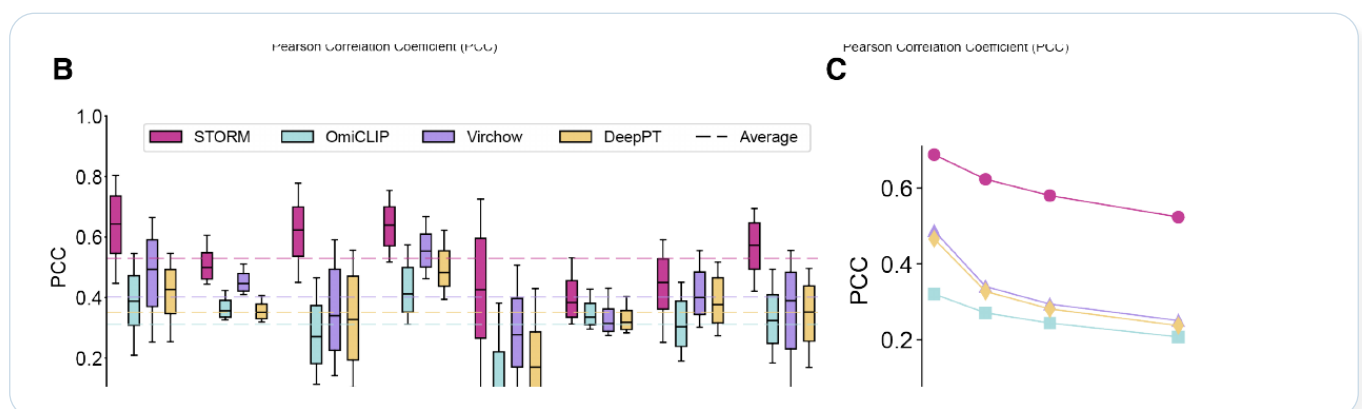


Figure 6: Figure 3B-C of the paper. STORM achieves better top-gene prediction across tissues and maintains higher PCC as the number of evaluated genes increases.

FIGURE NOTE 5.5 Interpretation

This result is useful because H&E-to-ST evaluation can depend heavily on which genes are selected. Showing stronger performance over different gene-panel sizes makes the virtual ST claim more robust.

5.5 Clinically relevant biomarker maps

The paper visualizes H&E images, measured ST, STORM predictions, and Virchow predictions for clinically relevant biomarkers including FGFR3, ERBB2, BCL2L1, CDK4, MET, COL1A1, EGFR, and KLK2. STORM produces spatial patterns with much higher correlation to true ST in several cancers.

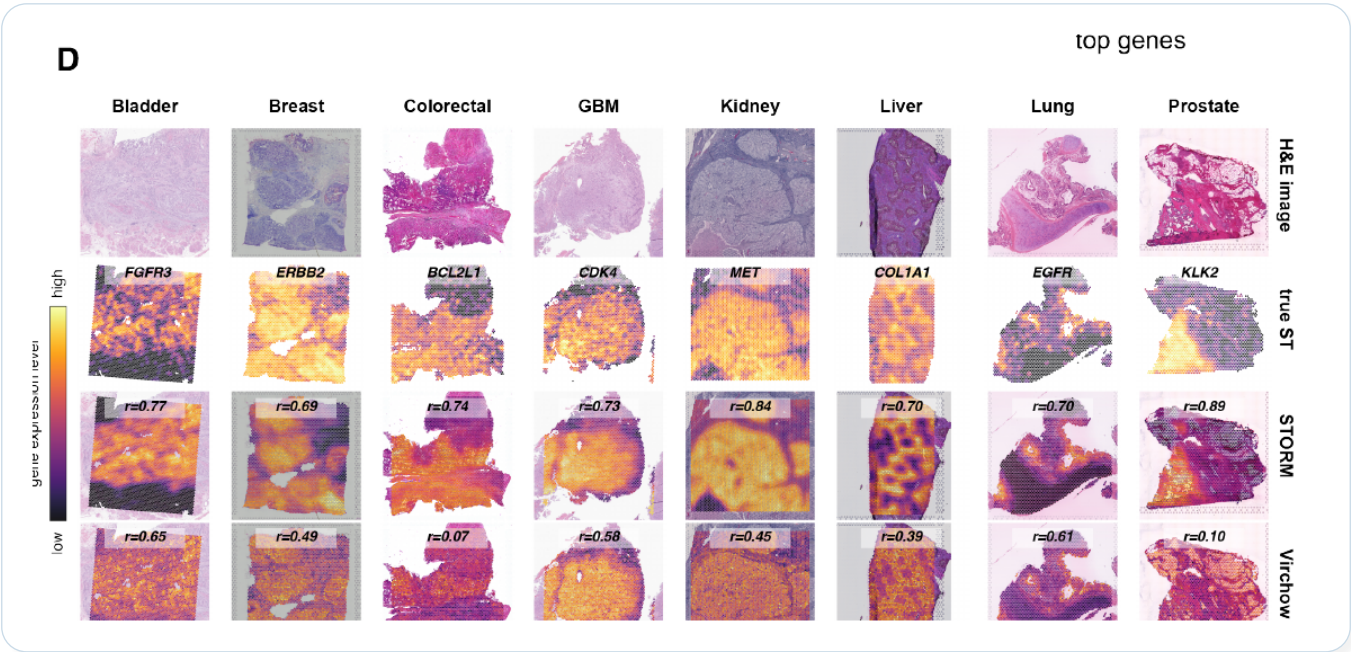


Figure 7: Figure 3D of the paper. Visual examples of virtual ST prediction for clinically relevant genes. STORM better recovers spatial expression patterns than Virchow in the displayed tumor types.

FIGURE NOTE 5.6 What this adds

The benchmark numbers are important, but the maps are more intuitive. They show whether predicted expression preserves spatial structure, not just aggregate correlation. This is critical if virtual ST is meant to support biological interpretation.

5.6 Clinical prediction workflow and cohorts

After generating virtual ST from H&E, STORM integrates H&E features and virtual ST features through cross-attention and AbMIL for patient-level outcome prediction. The paper evaluates colorectal and lung cancer cohorts with external validation sets.

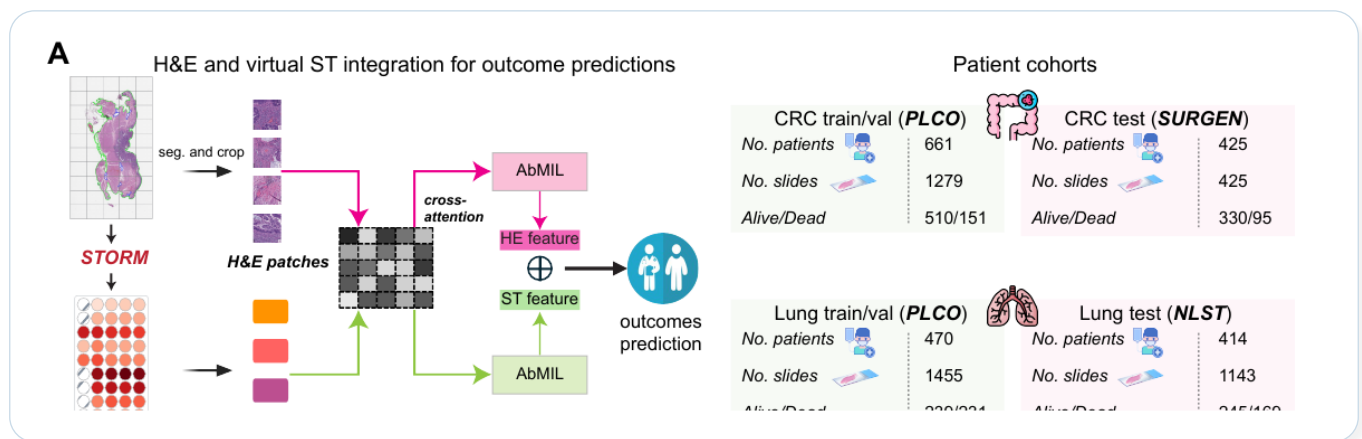


Figure 8: Figure 4A of the paper. Clinical prediction workflow. STORM generates virtual ST from routine H&E slides, encodes H&E and virtual ST features, integrates them through cross-attention, and predicts patient outcomes with AbMIL.

FIGURE NOTE 5.7 Why this design is clinically attractive

The deployment setting does not require real ST. STORM uses expensive ST during pretraining and then converts routine H&E slides into molecularly informed representations for prognosis prediction.

5.7 Prognosis prediction: C-index comparison

STORM is compared with DeepPT, OmiCLIP, H&E-only, and virtual-ST-only models. Across colorectal and lung validation/test cohorts, STORM achieves the highest C-index, with reported improvements over unimodal and multimodal baselines.

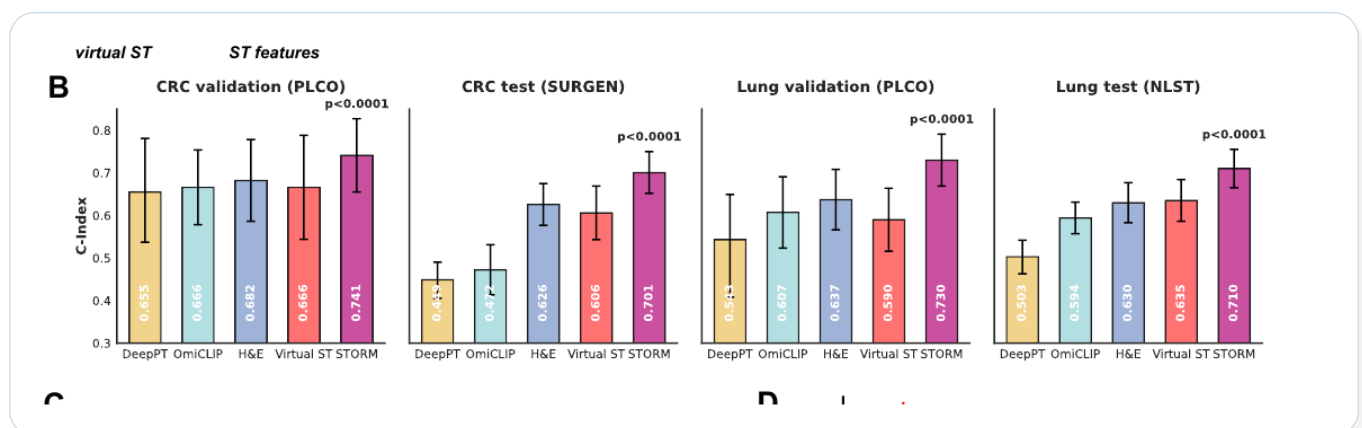


Figure 9: Figure 4B of the paper. C-index comparisons for colorectal and lung prognosis prediction. STORM outperforms unimodal H&E, virtual ST, DeepPT, and OmiCLIP baselines across validation and external test cohorts.

FIGURE NOTE 5.8 Interpretation

This supports the clinical value of combining morphology and virtual molecular information. The virtual ST branch alone is informative, but the best performance comes from multimodal integration.

5.8 Risk stratification with Kaplan-Meier curves

The paper applies fixed risk-score thresholds from validation cohorts to test cohorts and shows separation between high- and low-risk patients. The hazard ratios remain substantial across colorectal and lung valida-

tion/test cohorts.

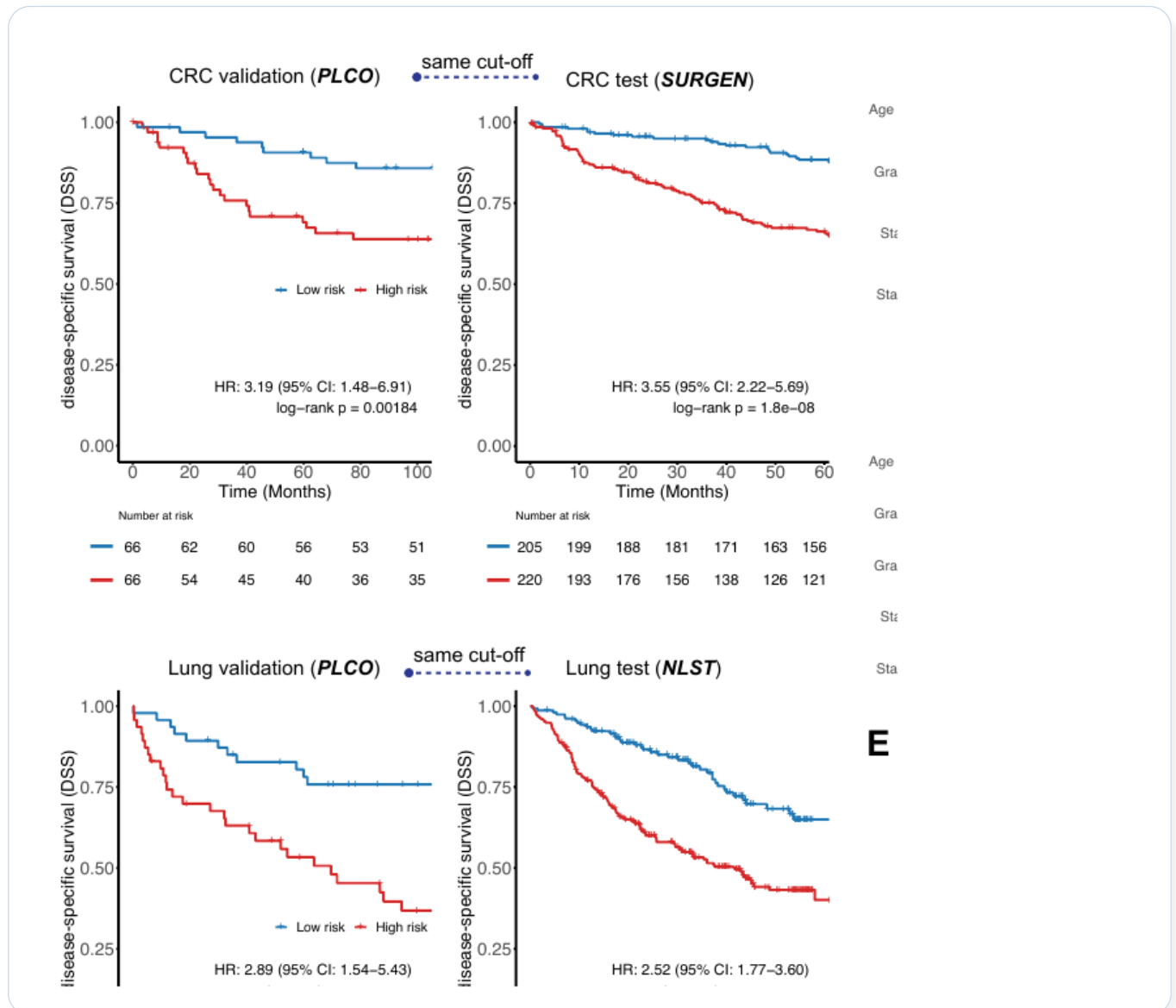


Figure 10: Figure 4C of the paper. Kaplan-Meier curves show that STORM stratifies patients into clinically meaningful high- and low-risk groups in colorectal and lung cancer cohorts.

FIGURE NOTE 5.9 Key caution

The survival curves are compelling, but they should be interpreted alongside cohort design, censoring, and clinical covariates. The most convincing part is that the same cutoff is transferred to independent test cohorts rather than tuned separately on each test set.

5.9 Independent prognostic value beyond clinical variables

STORM remains an independent prognostic factor after adjusting for age, grade, and stage. The paper also shows that adding STORM to clinical variables improves the C-index compared with grade or stage alone.

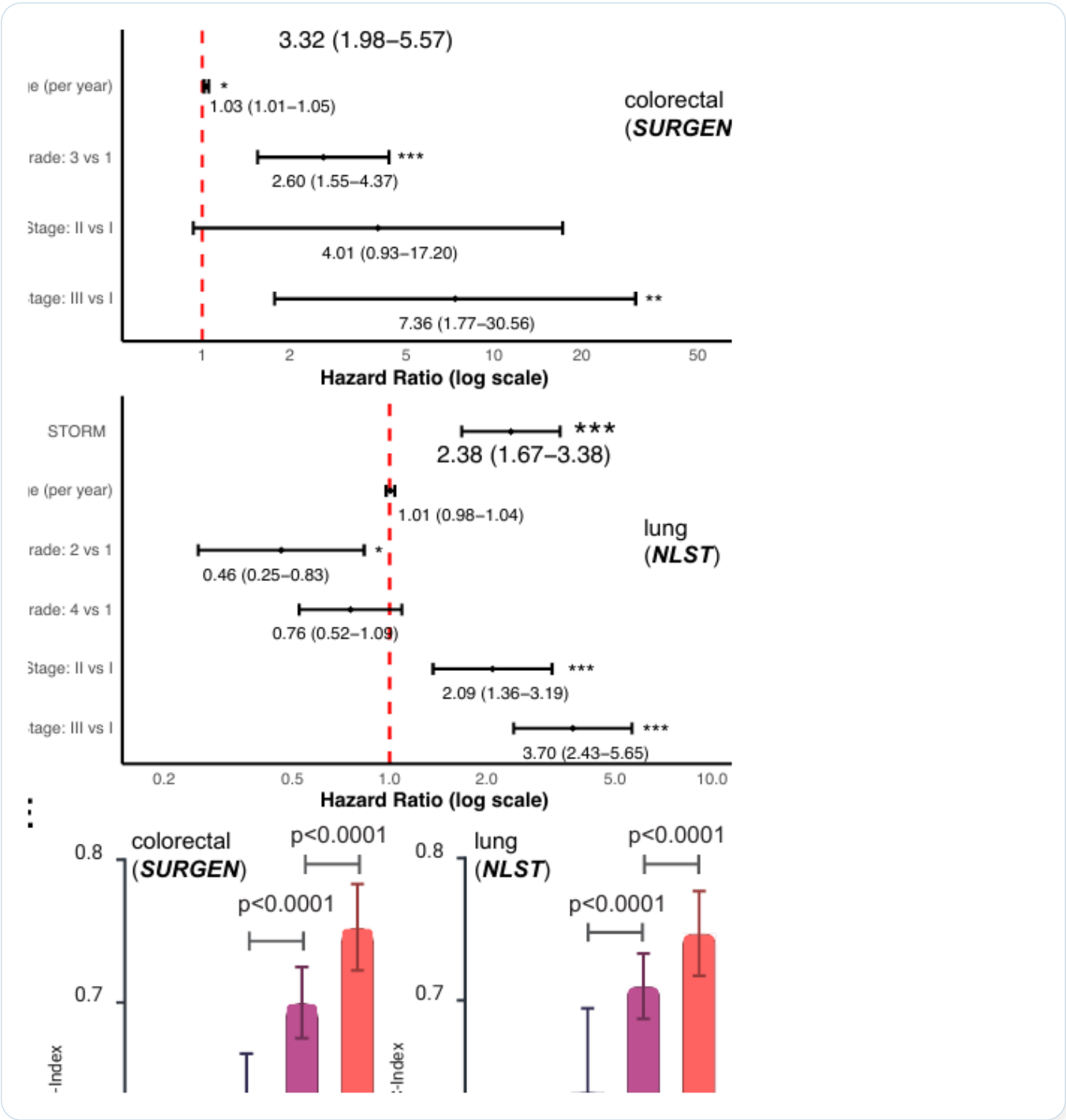


Figure 11: Figure 4D-E of the paper. Multivariate Cox analysis and clinical-variable comparison. STORM provides prognostic information beyond established clinical factors, and STORM plus clinical variables yields the strongest prediction.

FIGURE NOTE 5.10 Clinical implication

This is the strongest translational argument. If a model only repeats tumor stage, it has limited value. Here, the authors argue that STORM captures additional spatial-molecular information not fully explained by clinical variables.

5.10 Tissue-level interpretation of prognosis

The paper uses IG-derived risk maps and pathologist review to interpret high- and low-risk regions in colorectal and lung cancer. High-risk colorectal regions show adverse morphologies such as micropapillary subtype,

high-grade tumor budding, and infiltrative growth; low-risk regions show stronger lymphocytic infiltration. Lung examples similarly separate high-risk STAS/necrosis-like regions from lower-risk immune-rich or better differentiated regions.

A. tissue-level interpretation

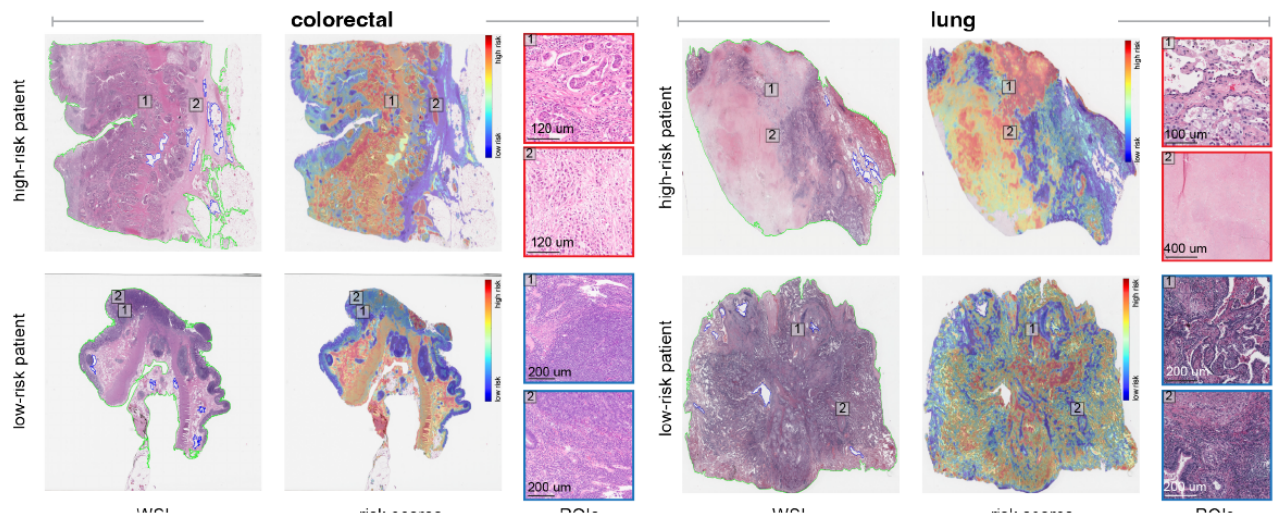


Figure 12: Figure 5A of the paper. Tissue-level interpretation. STORM risk maps highlight high- and low-risk regions in colorectal and lung cancer that correspond to recognizable pathological patterns.

FIGURE NOTE 5.11 Why this matters

This figure is important because it translates model risk into histological evidence. It helps move the work from black-box prediction toward pathologist-interpretable morphology.

5.11 Cell-level interpretation of prognosis

The authors quantify cell-type composition in high- and low-risk regions. High-risk regions are enriched for cancer cells, fibroblasts, and endothelial cells, while low-risk regions contain more lymphocytes and plasma cells.

B. cell-level interpretation

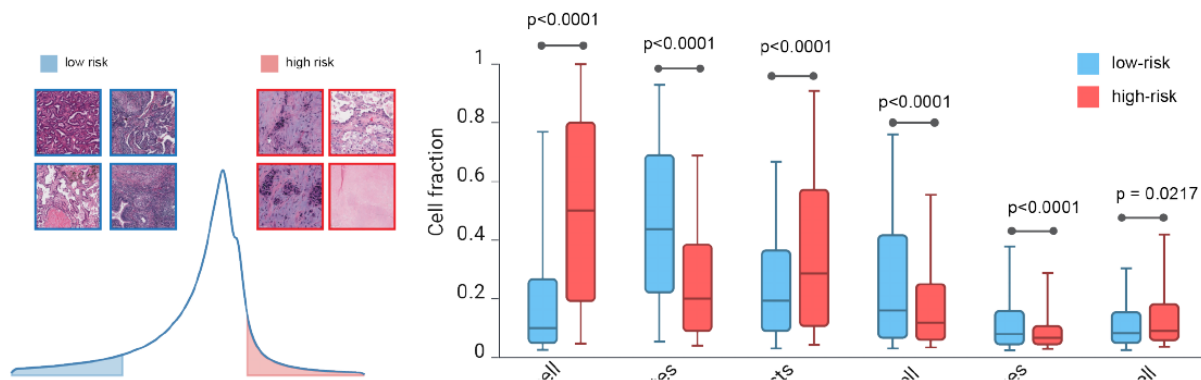


Figure 13: Figure 5B of the paper. Cell-level interpretation. Cell composition differs between high- and low-risk regions, linking STORM risk to tumor burden, stromal remodeling, vascular components, and immune infiltration.

FIGURE NOTE 5.12 My interpretation

This is a helpful bridge between image-level attention/risk maps and biological mechanisms. The model's risk signal is not only a heatmap; it can be decomposed into tumor, stromal, vascular, and immune components.

5.12 Gene-level interpretation of prognosis

Finally, the paper computes gene-level risk attributions and enrichment pathways. The identified genes include CD74, HLA-A/C, B2M, HSP90AA1/AB1, EPAS1, EZR, and XPO1. Enriched pathways include interferon-gamma response, unfolded protein response, cell-cycle and immune/stress pathways.

C. gene-level interpretation

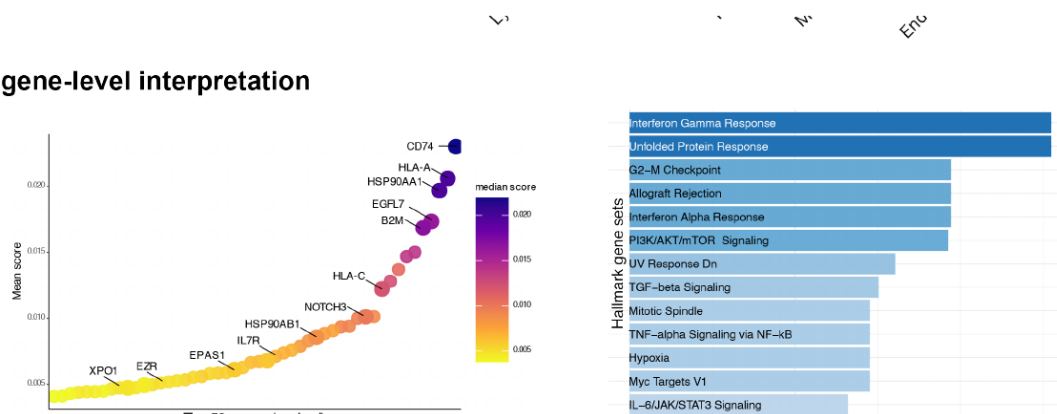


Figure 14: Figure 5C of the paper. Gene-level interpretation. STORM identifies prognostic genes and enriched pathways related to immune response, stress response, proliferation, and tumor signaling.

FIGURE NOTE 5.13 Interpretation

This final layer connects the model's clinical predictions back to virtual molecular programs. It is also where the method most clearly differs from H&E-only prognosis models: risk can be analyzed through inferred spatial gene-expression patterns.

6 What I learned

6.1 1. Spatial context is not optional for H&E-ST modeling

The paper’s strongest methodological message is that local tissue organization matters. A spot-level regression model can miss the structured neighborhood information that defines many biological compartments.

6.2 2. Virtual ST is useful only if it preserves spatial structure

Correlation numbers are not enough. The clinically relevant biomarker maps show whether predicted expression forms realistic spatial patterns. For biological discovery, spatial fidelity may matter as much as average PCC.

6.3 3. Clinical prediction becomes more interpretable when virtual ST is included

A pure H&E model may predict prognosis, but it is hard to explain mechanistically. STORM’s virtual ST branch allows interpretation at tissue, cell, and gene levels, making the clinical model easier to connect to biology.

7 Relationship to OmiCLIP and Multi-Embed

Table 2: How STORM relates to nearby H&E-omics models.

Model	Main input	Core idea	What STORM changes
OmiCLIP	H&E + ST gene sentences	Contrastive visual-transcriptomic alignment	STORM adds explicit spatial-neighborhood modeling and masked multimodal reconstruction.
Multi-Embed	H&E + multilayer omics / spatial omics	Unified morphology-molecule embedding across scales and omics layers	STORM focuses more deeply on spatial transcriptomics and virtual ST for clinical deployment.
PORPOISE	H&E + bulk omics	Task-specific multi-modal prognosis fusion	STORM generates spatially resolved virtual omics from H&E before outcome prediction.

STORM is closest to OmiCLIP in theme, but its core novelty is the spatially aware architecture. Compared with Multi-Embed, STORM is more specialized for H&E-ST representation learning and clinical virtual-ST applications.

8 Critical reading

CRITIQUE 8.1 Critical reading

Strengths. The paper has a strong foundation-model framing, large pretraining scale, explicit spatial architecture, broad virtual-ST benchmarks, clinical outcome evaluation, and multi-level biological interpretation.

Limitations. The manuscript is a work in progress, and some clinical claims would benefit from more detailed cohort descriptions, calibration analysis, and prospective validation. Virtual ST prediction may also reproduce correlations learned from training data rather than causal molecular programs.

Evaluation concern. It would be useful to see more ablations isolating the effects of the spatial encoder, neighborhood size, masked pretraining objective, H&E encoder choice, ST encoder choice, and virtual-ST quality on downstream prognosis.

9 Reusable ideas for my own research

1. **Spatially aware multimodal pretraining:** incorporate neighborhood context when aligning histology and molecular profiles.
2. **Virtual omics as an intermediate representation:** use H&E-derived molecular maps as interpretable bridges for clinical prediction.
3. **Domain discovery plus clinical validation:** show that discovered spatial domains are linked to genes, pathways, and survival.
4. **Multi-level interpretation:** connect prediction to tissue morphology, cell composition, and gene/pathway attribution.

10 Open questions

QUESTION 10.1 Open questions

- ▶ How much of STORM's gain comes from spatial context versus larger pretraining scale?
- ▶ Does virtual ST remain reliable for rare tissues, rare tumor subtypes, or strong staining/domain shifts?
- ▶ How stable are the gene-level risk attributions under different baselines or attribution methods?
- ▶ Can STORM be adapted to partially paired or unpaired histology-omics datasets?
- ▶ In clinical prediction, does virtual ST improve calibration and decision utility, or mainly ranking metrics such as C-index?

11 Final takeaway

STORM is a strong example of the next step in pathology-omics modeling: not just aligning H&E and molecular data, but building a spatially aware foundation model that can generate virtual ST and use it for interpretable clinical prediction. Its most reusable idea is the combination of neighborhood-aware H&E-ST pretraining with downstream biological and clinical validation.